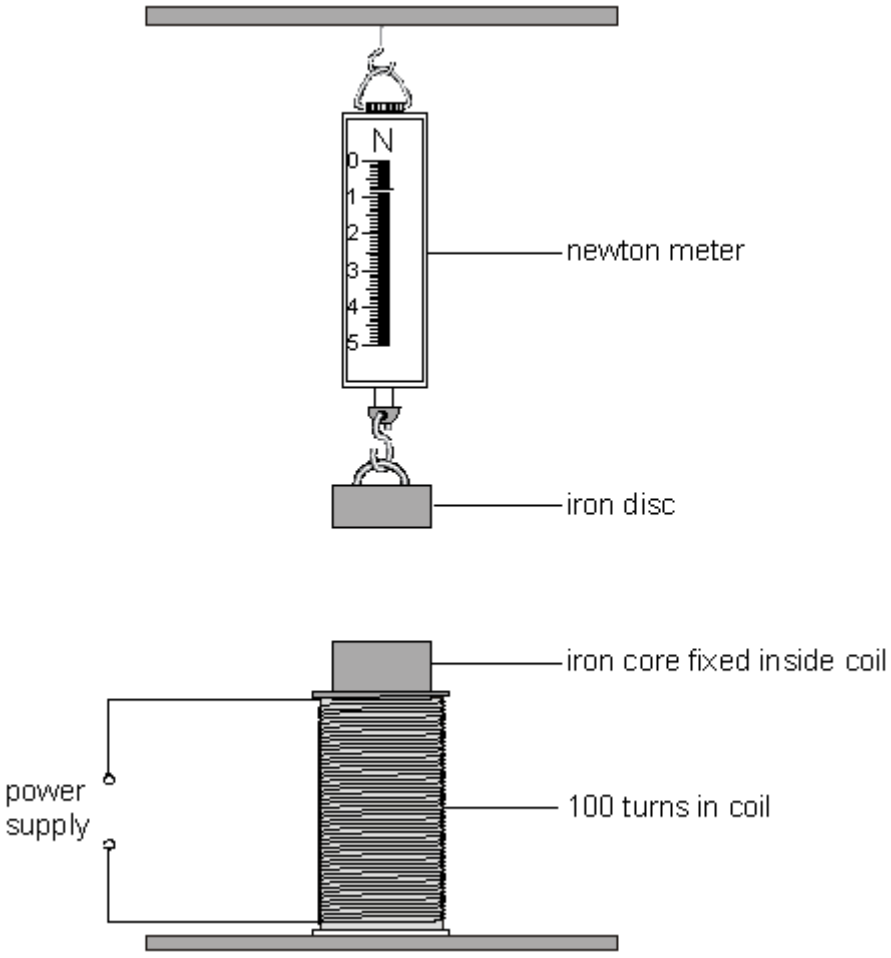
**Q1.** Mary used the apparatus below to test the strength of an electromagnet.
She used the reading on the newton meter to measure the force of the magnet on the iron disc.




(a) Explain why the reading on the newton meter increases when a current passes through the coil.

.....

.....

.....

.....

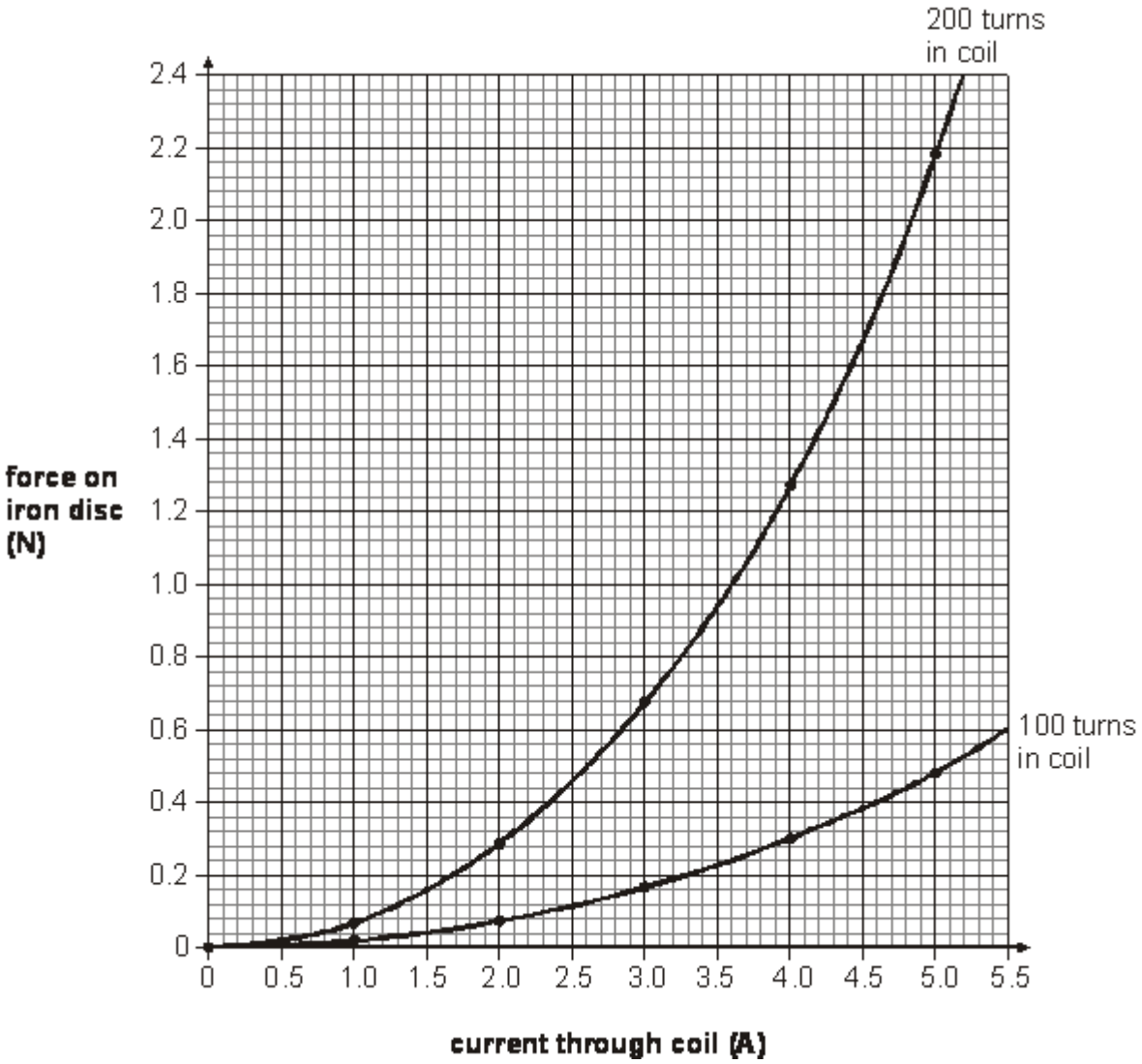
2 marks

(b) When a current passes through the coil, some of the electrical energy is changed to thermal energy.
What would happen to the coil if the current passing through it was too large?

.....

1 mark

(c) Mary made two electromagnets, one with 100 turns of wire in the coil and one with 200 turns.
She varied the current through the coil of each electromagnet.
She measured the force of each electromagnet on the iron disc.
The graph shows her results.




Write **two** conclusions that Mary could make from these results.

1. .....

.....

2. .....

.....

2 marks
maximum 5 marks

Mark schemes

## Q1.

(a) • the core becomes magnetised

*accept 'the disc becomes magnetised'*
*accept 'there is a magnetic field'*
*accept 'it becomes an (electro)magnet'*
*accept 'the stronger the current the stronger the magnet'*

*do **not** accept 'the iron core becomes magnetic'*

*do **not** accept 'the magnet gets stronger'*

**1 (L7)**

any **one** from

• the core attracts the iron disc

*accept 'the disc is pulled down'*

*'the disc moves down' is insufficient as it does not imply that a force is exerted*

• there is more force on the iron disc

*accept 'the magnet exerts a force on the disc'*

*'there is a force on the disc' is insufficient as it does not refer to the origin of the force*

**1 (L7)**

(b) any **one** from

• it would melt

*accept 'it would fuse'*

*'the coil would break' is insufficient*

• it would get too hot

*accept 'it could catch fire'*
*accept 'it would blow'*

*'it would get hot' is insufficient*

**1 (L7)**

(c) • the greater the current, the greater the force **or** field

*accept the converse*

*answers must refer to a pattern describing a continuous variable*
*do **not** accept 'it becomes more magnetic'*

• the more turns, the greater the force **or** field

*accept the converse*

*accept 'the more turns, the more powerful **or** stronger the magnet'*

*answers must include a comparison*
*'the more turns, the more powerful it is' is insufficient*

*accept 'the electromagnet with 200 turns is stronger'*
*accept 'doubling the turns more than doubles the force'*

*award one mark if the answer refers to a number of coils rather than number of turns*

**2 (L7)**

**[5]**